Technical School of
Telecommunications Engineering
and Systems

ANTENNAS AND EMC

SOLUTION

LABORATORY PRACTICE 3
MICROSTRIP ANTENNAS

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This laboratory exercise designs and simulates a rectangular microstrip antenna fed by a coaxial probe. Digits from the assigned identification number determine the operating frequency and, for the second design, the dielectric-substrate properties.

The goal is to compare two antenna versions. The first uses FR4, which has moderate relative permittivity and appreciable loss. The second uses a thicker, low-permittivity dielectric. The comparison shows how the substrate affects the reflection coefficient, impedance bandwidth, and radiation patterns.

Both designs follow the same procedure. The patch geometry and feed position are first estimated analytically. The principal dimensions and probe location are then swept in the electromagnetic model until the best match is obtained at the assigned frequency.

DNI: 02757023*J*

$C_8 = 0,\ C_7 = 2,\ C_6 = 7,\ C_5 = 5,\ C_4 = 7,\ C_3 = 0,\ C_2 = 2,\ C_1 = 3$

$f = 1500 + 100C_3 + 10C_2 + C_1 = 1500 + 0 + 20 + 3 = 1523$ MHz

$f = 1{,}523$

2. DESIGN 1 - FR4 MICROSTRIP ANTENNA

• Patch sizing:

1. Approximate optimum patch width:

$$a = c0/(2f) \cdot \sqrt{(2/(\varepsilon r + 1))}$$

$$\mathbf{a = 300/(2 \cdot 1,523) \cdot \sqrt{(2/(4,5 + 1)) = 59,39\ mm}}$$

2. Effective dielectric constant:

$$\varepsilon eff = (\varepsilon r + 1)/2 + (\varepsilon r - 1)/2 \cdot (1 + 12h/a)^{(-1/2)}$$

$$\varepsilon eff = 5,5/2 + 3,5/2 \cdot (1 + 12 \cdot 1,5/59,39)^{(-1/2)}$$

$$\mathbf{\varepsilon eff = 2,75 + 1,75 \cdot (1,303)^{(-0,5)} = 2,75 + 1,75 \cdot 0,876 \approx 4,283}$$

3. Fringing-field length extension:

$$(\Delta b/2)/h = 0,412 \cdot [(\varepsilon eff + 0,3)(a/h + 0,264)] / [(\varepsilon eff - 0,258)(a/h + 0,8)]$$

Substituting $\varepsilon eff = 4,283$ and $a/h = 39,59$:

$$(\Delta b/2)/1,5 = 0,412 \cdot [(4,583)(39,86)] / [(4,025)(40,39)] = 0,412 \cdot 1,124 \approx 0,463$$

$$\Delta b/2 = 0,463 \cdot 1,5 = 0,694\ mm$$

$$\mathbf{\Delta btotal = 2 \cdot (\Delta b/2) \approx 1,39\ mm}$$

4. Final physical length:

$$b = c0/(2fr\sqrt{\varepsilon eff}) - \Delta btotal$$

$$\mathbf{b = 300/(2 \cdot 1,523 \cdot \sqrt{4,283}) - 1,39 = 47,59 - 1,39 \approx 46,20\ mm}$$

- **Feed-point estimate:**

1. Conductor losses:

$$Qc = h\sqrt{(\pi f \mu_0 \sigma)}$$

$$\mathbf{Qc = 1{,}5\cdot10^{-3}\sqrt{(\pi\cdot1{,}523\cdot10^{9}\cdot4\pi\cdot10^{-7}\cdot5{,}8\cdot10^{7})} \approx 886}$$

2. Dielectric losses:

$$\mathbf{Qd = 1/\tan\delta = 1/0{,}01 = 100}$$

3. Radiation quality factor:

$$Qr = c_0\sqrt{\varepsilon_{eff}}/(4hf) - \varepsilon_{eff}(\Delta b/2)/h \text{ Term 1:}$$
$$3\cdot10^{8}\sqrt{4{,}283}/(4\cdot0{,}0015\cdot1{,}523\cdot10^{9}) \approx 67{,}24$$

Term 2: $4{,}283\cdot0{,}000694/0{,}0015 \approx 1{,}98$

$$\mathbf{Qr \approx 67{,}24 - 1{,}98 = 65{,}9}$$

4. Total quality factor:

$$1/QT = 1/Qr + 1/Qc + 1/Qd = 1/65{,}9 + 1/886 + 1/100$$

$$\mathbf{1/QT \approx 0{,}0152 + 0{,}0011 + 0{,}0100 = 0{,}0263 \quad \Rightarrow \quad QT \approx 38{,}0}$$

5. Efficiency:

$$\mathbf{\eta = QT/Qr = 38{,}0/65{,}9 \approx 57{,}7\ \%}$$

6. Radiation resistance:

$Rrad = Qr\cdot2h/(2\pi f\varepsilon_0\varepsilon_{eff}ab)$ Substituting $Qr = 65{,}9$, $h = 1{,}5$ mm, $a = 59{,}39$ mm y $b = 46{,}20$ mm:

$$\mathbf{Rrad \approx 198{,}6\ \Omega}$$

7. Total resistance:

$$\mathbf{RT = Rrad\cdot\eta \approx 198{,}6\cdot0{,}577 \approx 114{,}6\ \Omega}$$

8. Feed-position calculation:

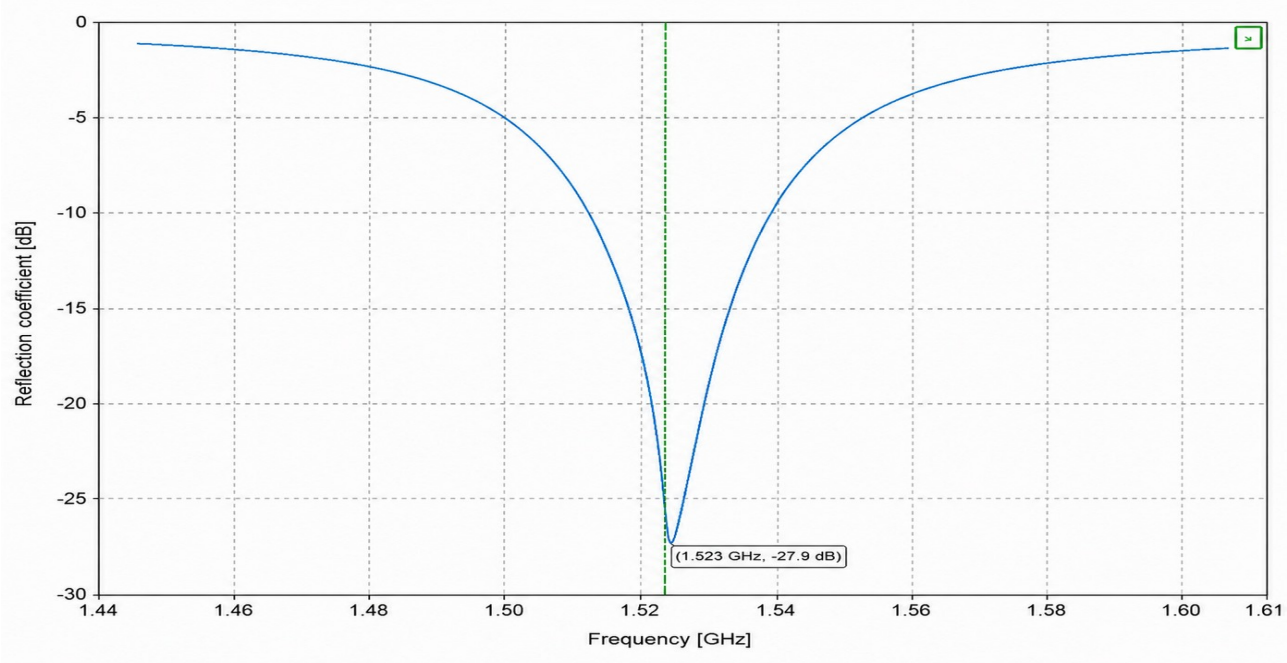
A 50-ohm match is estimated from the cosine-squared resistance variation measured from the patch center:

$$Rin(y) = RT \cdot \cos^2(\pi \cdot y/b)$$
$$50 = 114{,}6 \cdot \cos^2(\pi \cdot y/46{,}20)$$
$$\sqrt{(50/114{,}6)} = \cos(\pi \cdot y/46{,}20) \Rightarrow 0{,}661 = \cos(0{,}0680y)$$
$$0{,}0680y = \arccos(0{,}661) \approx 0{,}849 \text{ rad}$$
$$\mathbf{y = 12.49 \text{ mm}.}$$

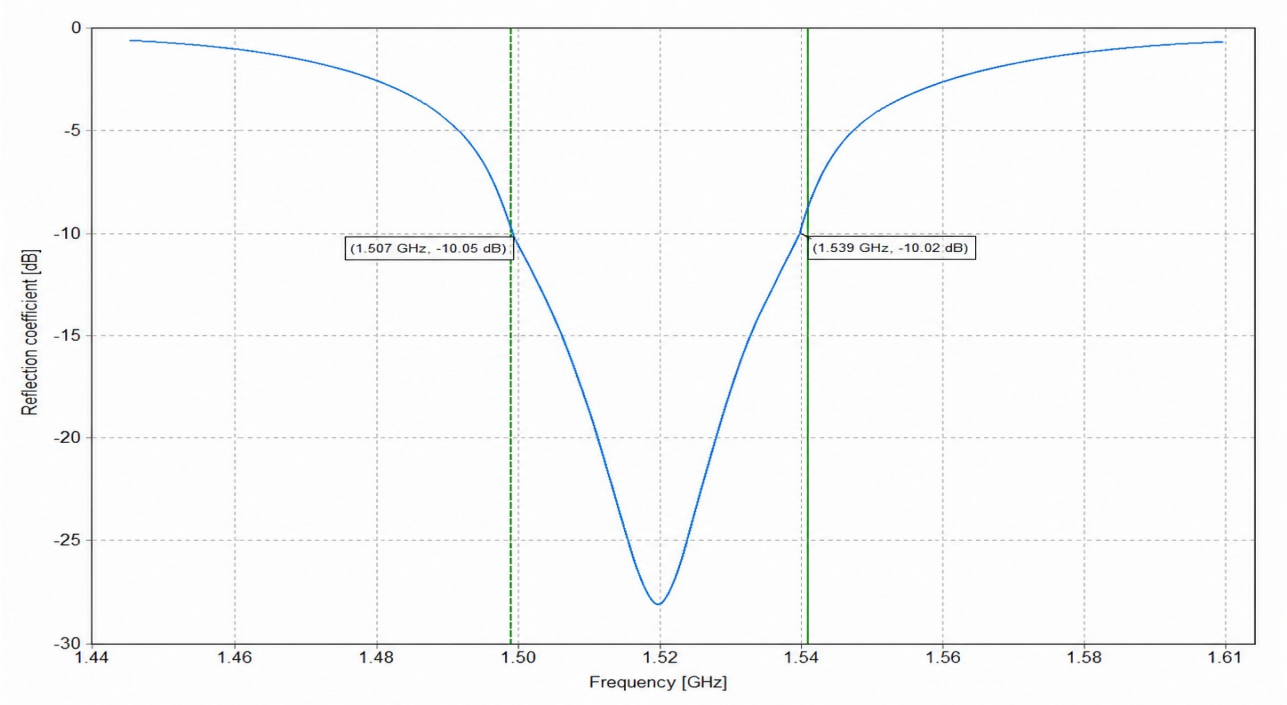
• Parameter-sweep tuning:

The primary tuning curve is S11 in decibels. The patch length is adjusted to center the resonance at 1.523 GHz, while the probe position is adjusted to minimize the reflected power.

Final simulation values:
- Patch length: 44.5 mm
- Feed position: 10.5 mm



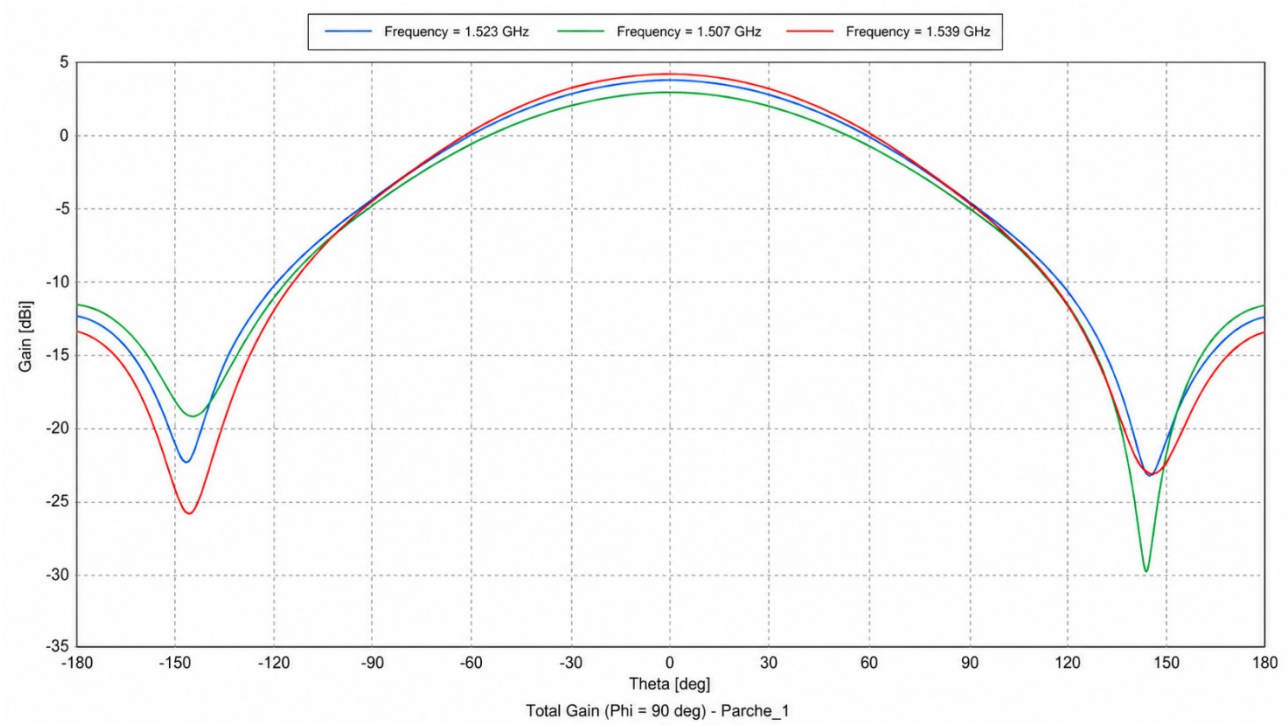
▯ Obtained bandwidth:

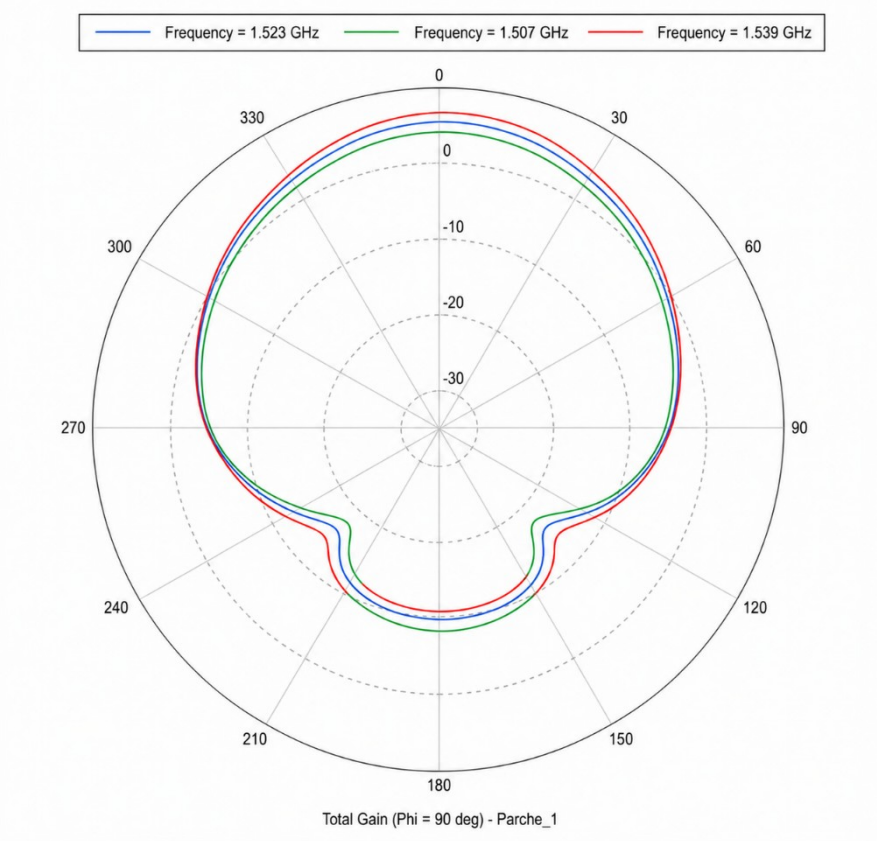


BW = 32.0 MHz

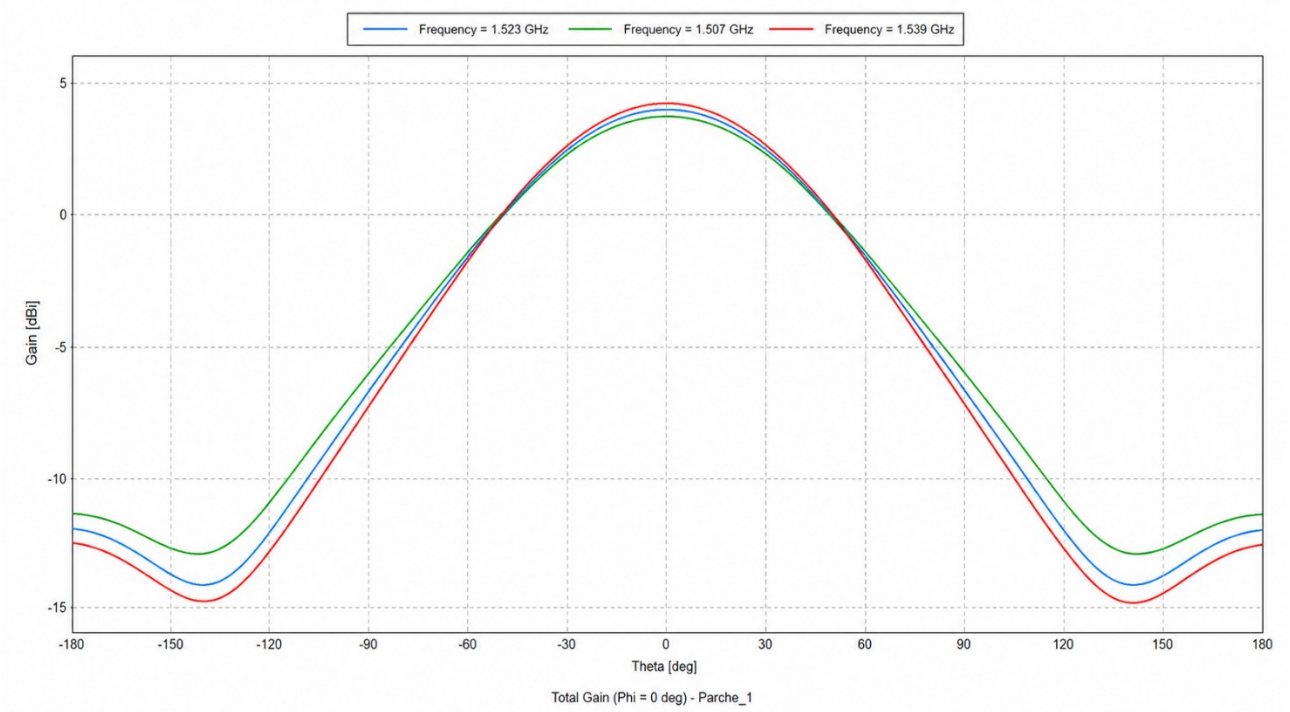
2.2. RADIATION PATTERNS: E- AND H-PLANES

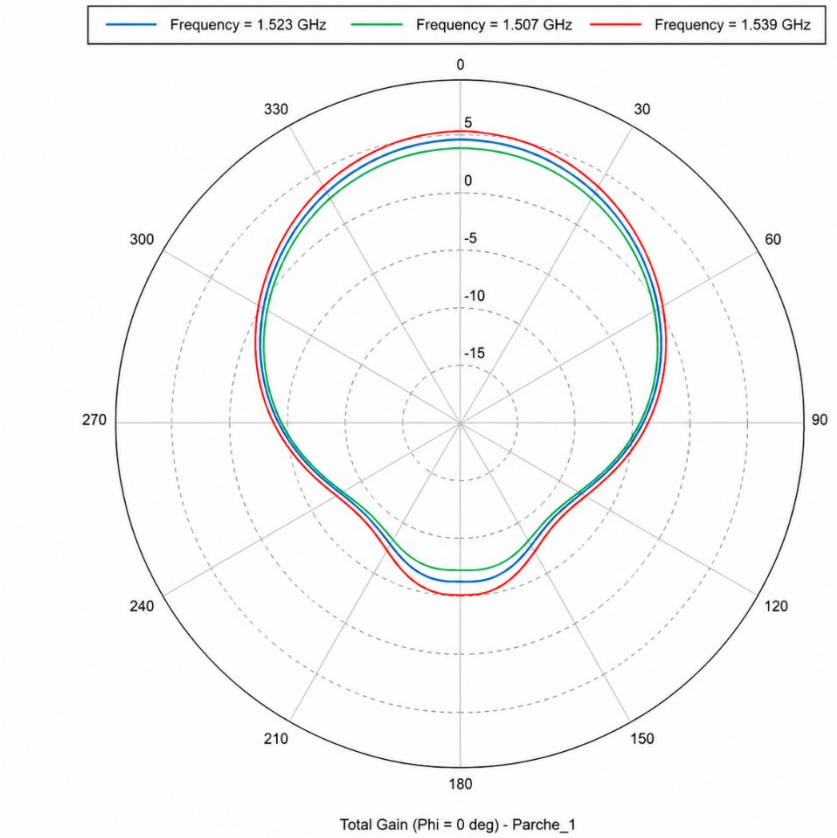
1. E-plane cut (electric-field plane)





2. H-plane cut (magnetic-field plane)





3. DESIGN 2 - LOW-PERMITTIVITY, THICKER SUBSTRATE

• Patch sizing:

1. Relative permittivity:

epsilon_r = 1.05 + 0.01C4 = 1.12.

tan(delta) = 0.0001(C5 + 1) = 0.0006.

h = λ0·(0,02 + 0,01·C6) λ0 = 300/1,523 = 196,98 mm h = 196,98·(0,02 + 0,01·7) = 196,98·0,09 ≈ 17,73 mm 4. Anchura aproximada óptima:

a = 300/(2·1,523) · √(2/(1,12 + 1)) = 95,66 mm 5. Effective dielectric constant:

εeff = (εr + 1)/2 + (εr − 1)/2 · (1 + 12h/a)^(-1/2) εeff = 2,12/2 + 0,12/2 · (1 + 12·17,73/95,66)^(-1/2) εeff = 1,06 + 0,06·(3,224)^(-0,5) = 1,06 + 0,06·0,557 ≈ 1,093 6. Fringing-field length extension:

(Δb/2)/h = 0,412·[(1,093+0,3)(5,396+0,264)] / [(1,093−0,258)(5,396+0,8)] ≈ 0,628

Δb/2 = 0,628·17,73 = 11,13 mm Δbtotal = 2·11,13 = 22,26 mm 7. Final physical length:

b = 300/(2·1,523·√1,093) − 22,26 = 94,19 − 22,26 ≈ 71,93 mm

- **Feed-point estimate:**

1. Conductor losses:

$Q_c = h\sqrt{(\pi f \mu_0 \sigma)}$ $Q_c = 17{,}73\cdot10^{-3}\sqrt{(\pi\cdot1{,}523\cdot10^{9}\cdot4\pi\cdot10^{-7}\cdot5{,}8\cdot10^{7})} \approx 10469$ 2. Dielectric losses:

$Q_d = 1/\tan\delta = 1/0{,}0006 \approx 1667$ 3. Radiation quality factor:

$$Q_r = c_0\sqrt{\varepsilon_{eff}}/(4hf) - \varepsilon_{eff}(\Delta b/2)/h$$ Term 1: $3\cdot10^{8}\sqrt{1{,}093}/(4\cdot0{,}01773\cdot1{,}523\cdot10^{9}) \approx 2{,}90$

Term 2: $1{,}093\cdot0{,}01113/0{,}01773 \approx 0{,}69$

$Q_r \approx 2{,}90 - 0{,}69 = 2{,}22$ 4. Total quality factor:

$$1/Q_T = 1/Q_r + 1/Q_c + 1/Q_d = 1/2{,}22 + 1/10469 + 1/1667 \approx 0{,}452$$

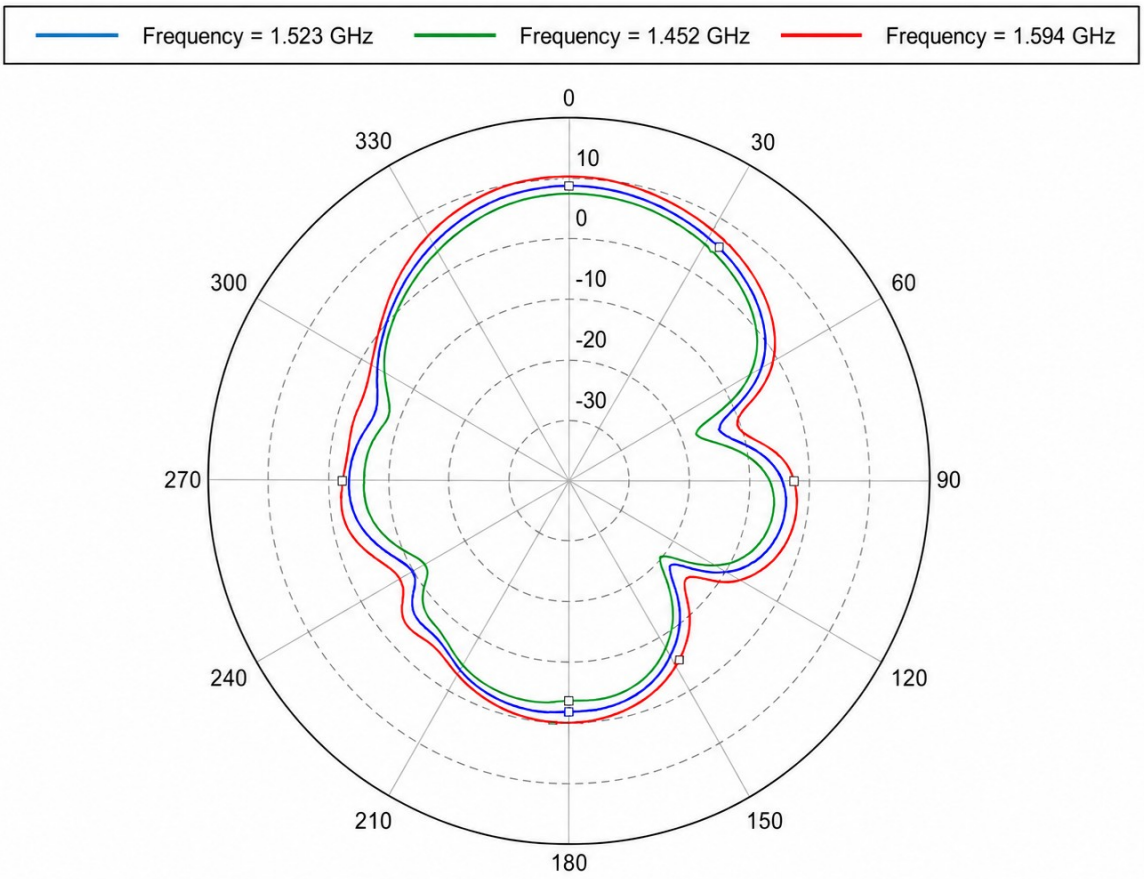
$Q_T \approx 2{,}21$ 5. Efficiency:

$\eta = Q_T/Q_r = 2{,}21/2{,}22 \approx 99{,}8\ \%$ 6. Radiation resistance:

$R_{rad} = Q_r\cdot2h/(2\pi f\varepsilon_0\varepsilon_{eff}ab)$ Substituting $Q_r = 2{,}22$, $h = 17{,}73$ mm, $a = 95{,}66$ mm y $b = 71{,}93$ mm:

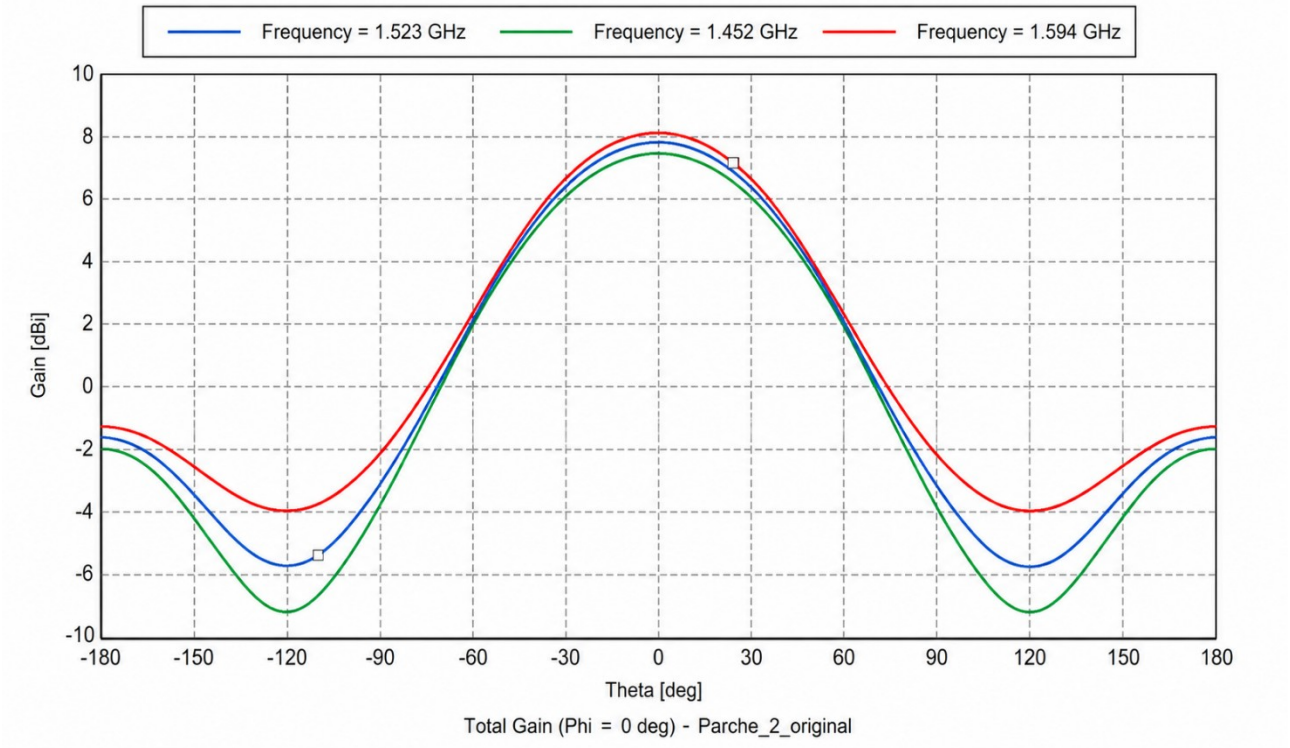
$R_{rad} \approx 123{,}3\ \Omega$ 7. Total resistance:

$$R_T = R_{rad}\cdot\eta \approx 123{,}3\cdot0{,}998 \approx 123{,}1\ \Omega$$

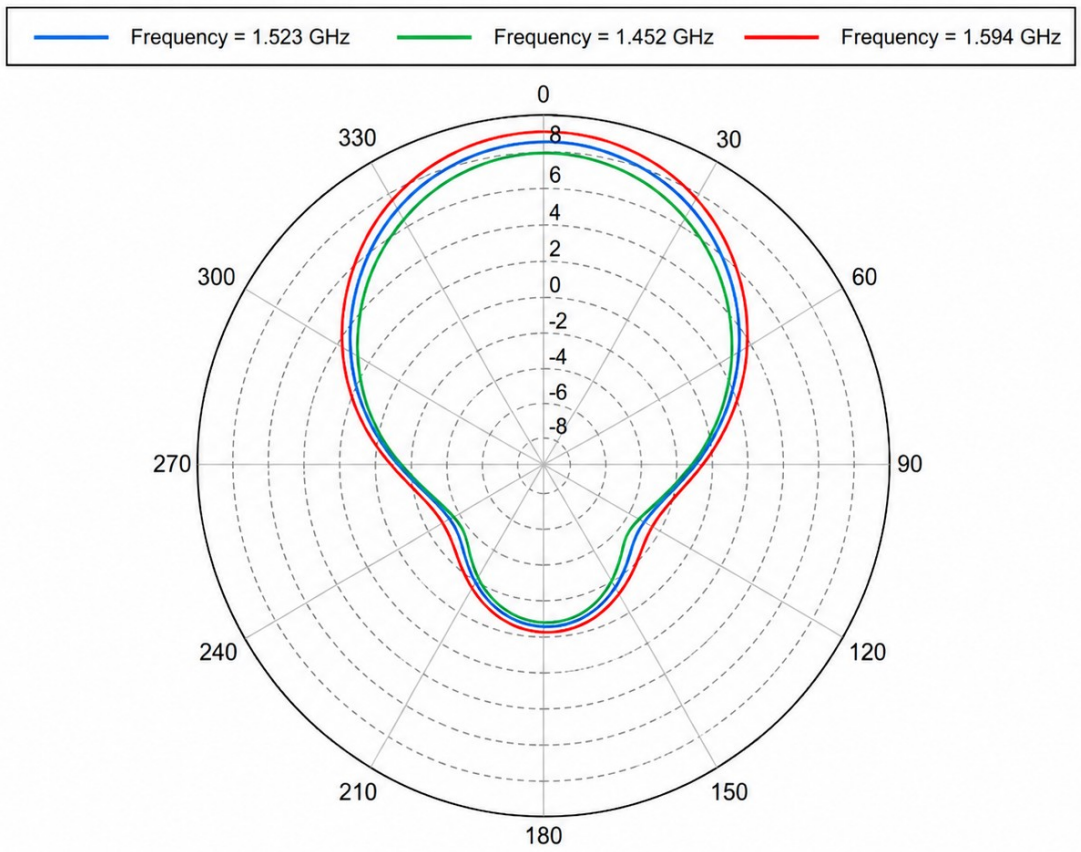


Total Gain (Phi = 90 deg) - Parche_2_original

2. H-plane cut (magnetic-field plane)



Total Gain (Phi = 0 deg) - Parche_2_original



Total Gain (Phi = 0 deg) - Parche_2_original

4.3. Technical Conclusions

The FR4 design is correctly matched at 1.523 GHz, with a reflection minimum near -28 dB and a 32.0 MHz bandwidth for S11 < -10 dB. Its calculated efficiency is lower because of dielectric loss, consistent with Qd = 100.

The thicker, low-permittivity design clearly increases the bandwidth to 142 MHz. Substrate loss is much lower, and the total quality factor is dominated by radiation. The calculated efficiency is therefore close to 100%.

Both designs radiate primarily in the broadside direction. The H-plane maintains a more regular main lobe, whereas the E-plane shows greater variation for the thicker design. In the comparison plots, the cross-polarized component remains below the co-polarized component, so the dominant patch polarization is the expected one.